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Lombardo

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(54) **LIFT DEVICE WITH INTERNALLY DEPLETABLE BATTERY**

(71) Applicant: **Oshkosh Corporation**, Oshkosh, WI (US)

(72) Inventor: **David Lombardo**, Oshkosh, WI (US)

(73) Assignee: **OSHKOSH CORPORATION**, Oshkosh, WI (US)

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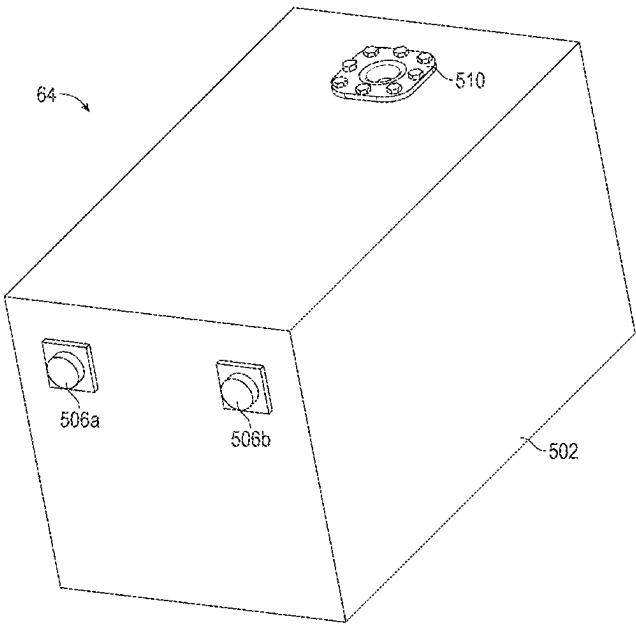
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Primary Examiner — Jonathan G Jelsma
(74) *Attorney, Agent, or Firm* — FOLEY & LARDNER LLP

(57) **ABSTRACT**

A battery pack for a lift device includes a housing, battery cells, a resistor, a conductive element, and a member. The resistor is electrically coupled with a positive terminal of the battery cells. The conductive element is configured to transition between an open state in which a discharge path is not defined between the positive terminal of the battery cells and a negative terminal or a ground, and a closed state in which the discharge path is defined. The member is disposed at the opening and accessible from an exterior of the housing. The member is manually transitionable by a technician between a first state in which the conductive element is in the open state, and a second state in which the conductive element is driven into the closed state such that the battery cells discharge remaining electrical energy via the discharge path.

13 Claims, 12 Drawing Sheets



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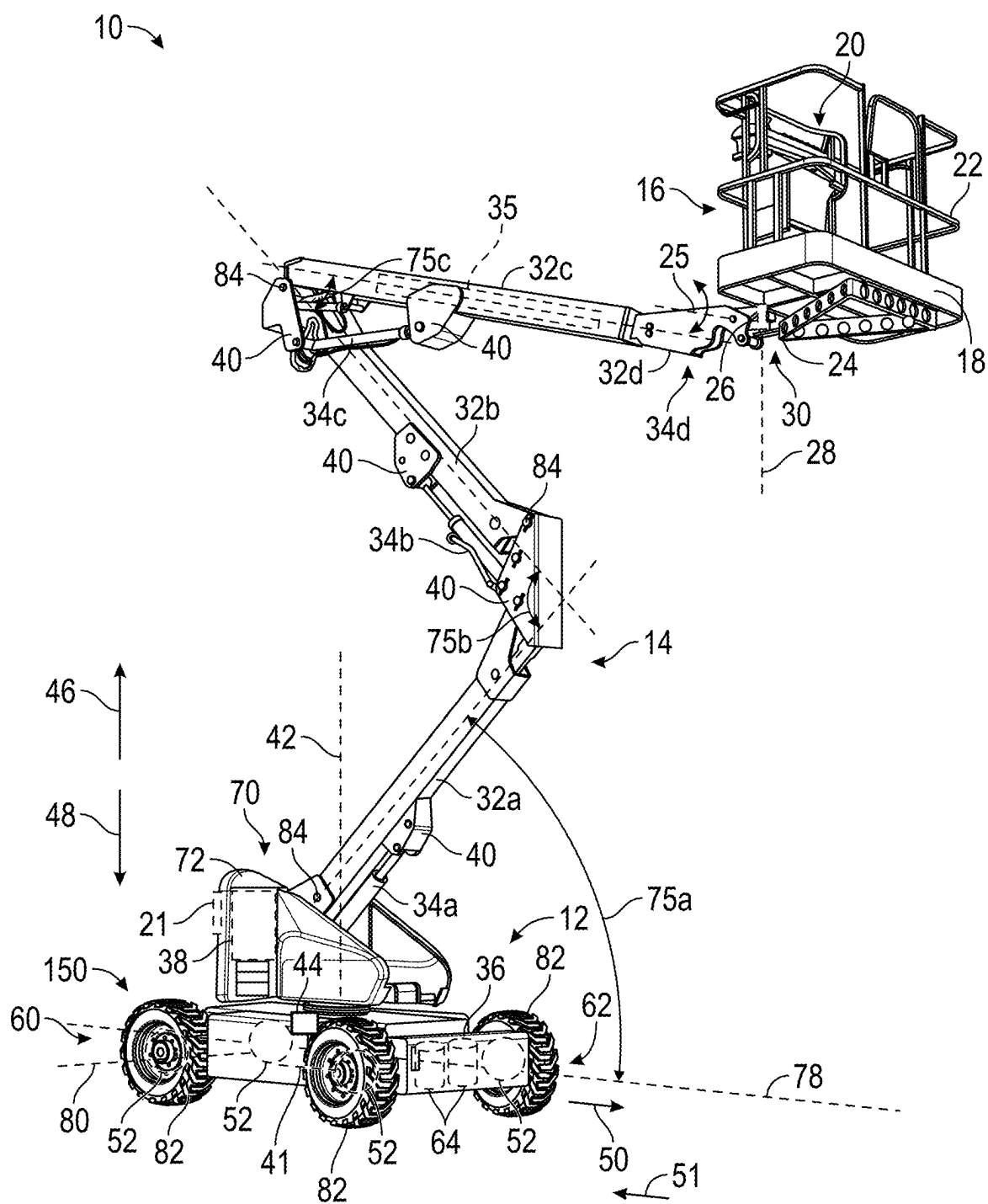


FIG. 1

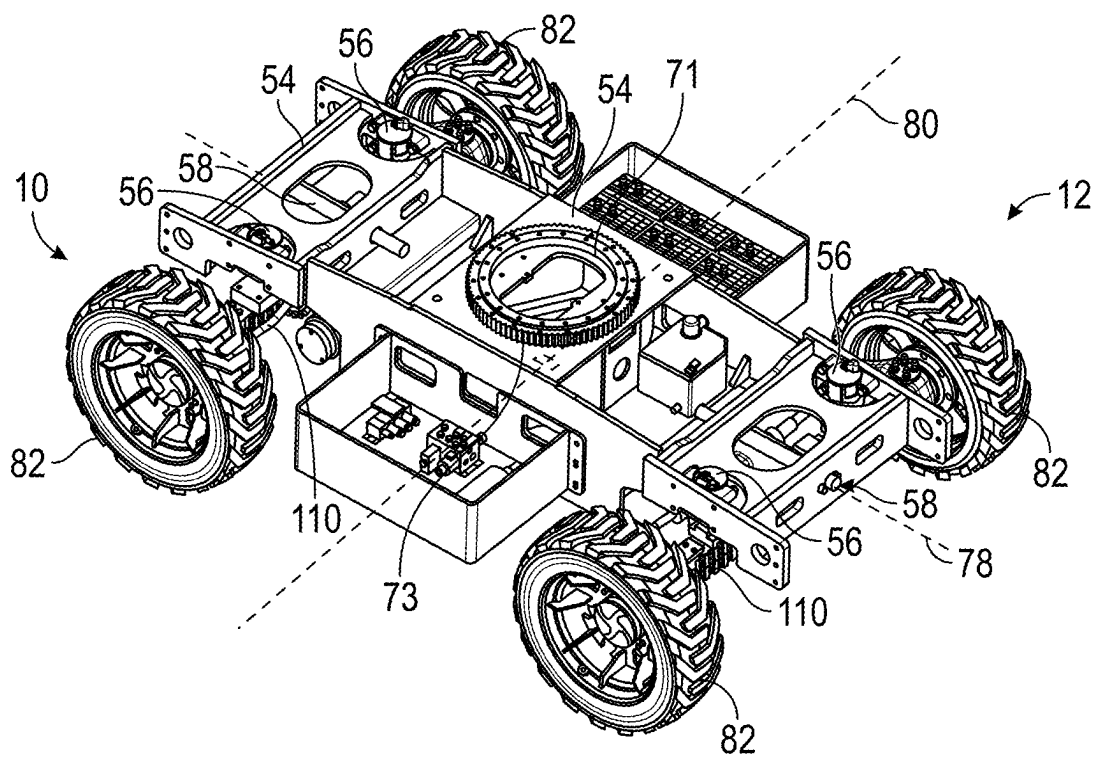


FIG. 2

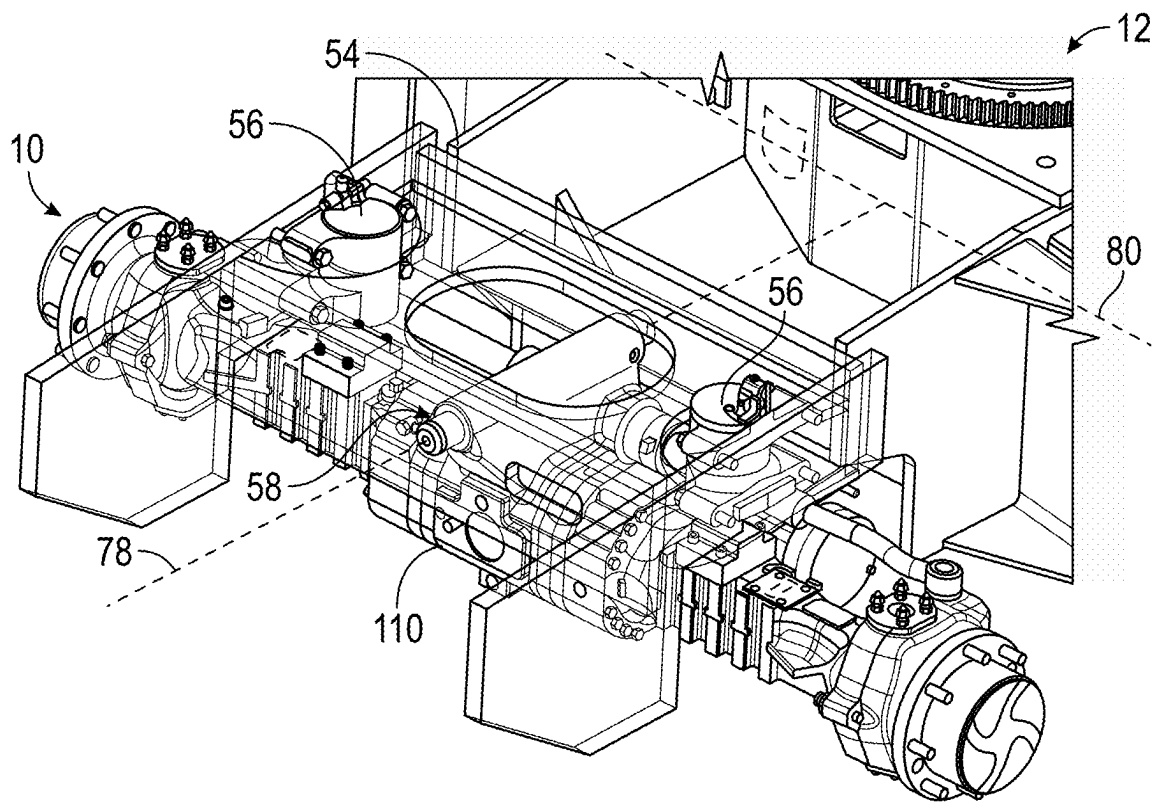


FIG. 3

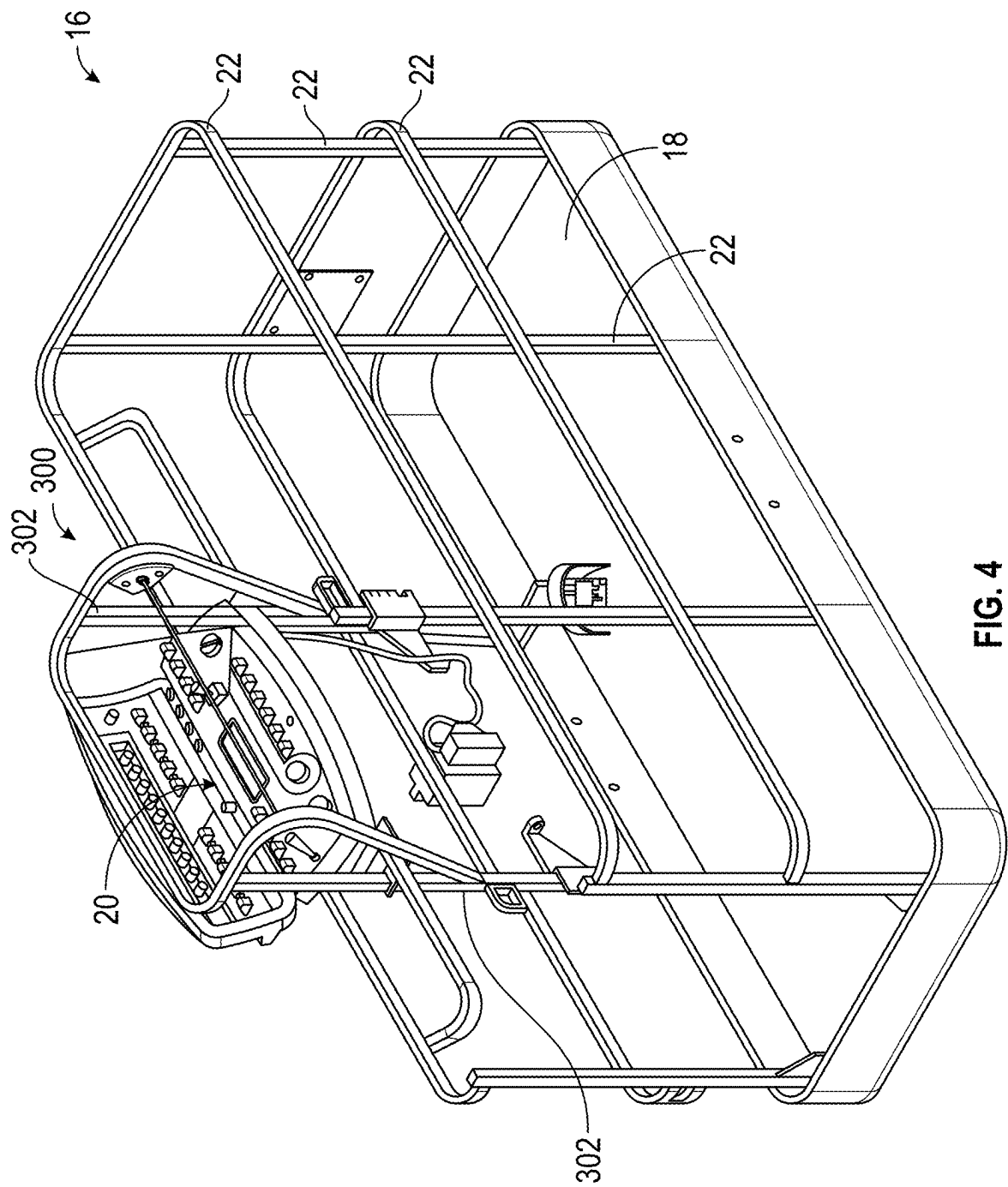


FIG. 4

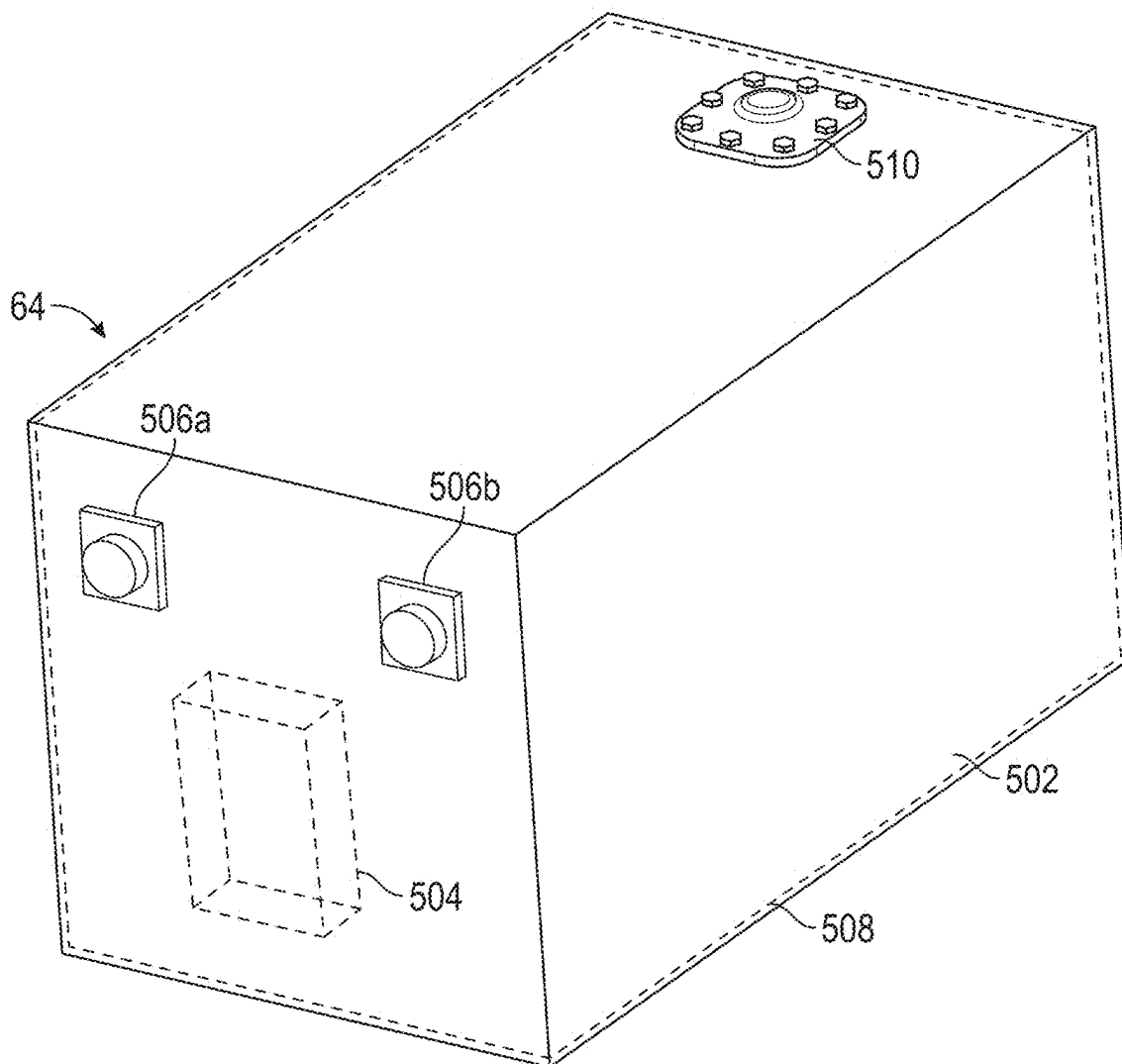


FIG. 5

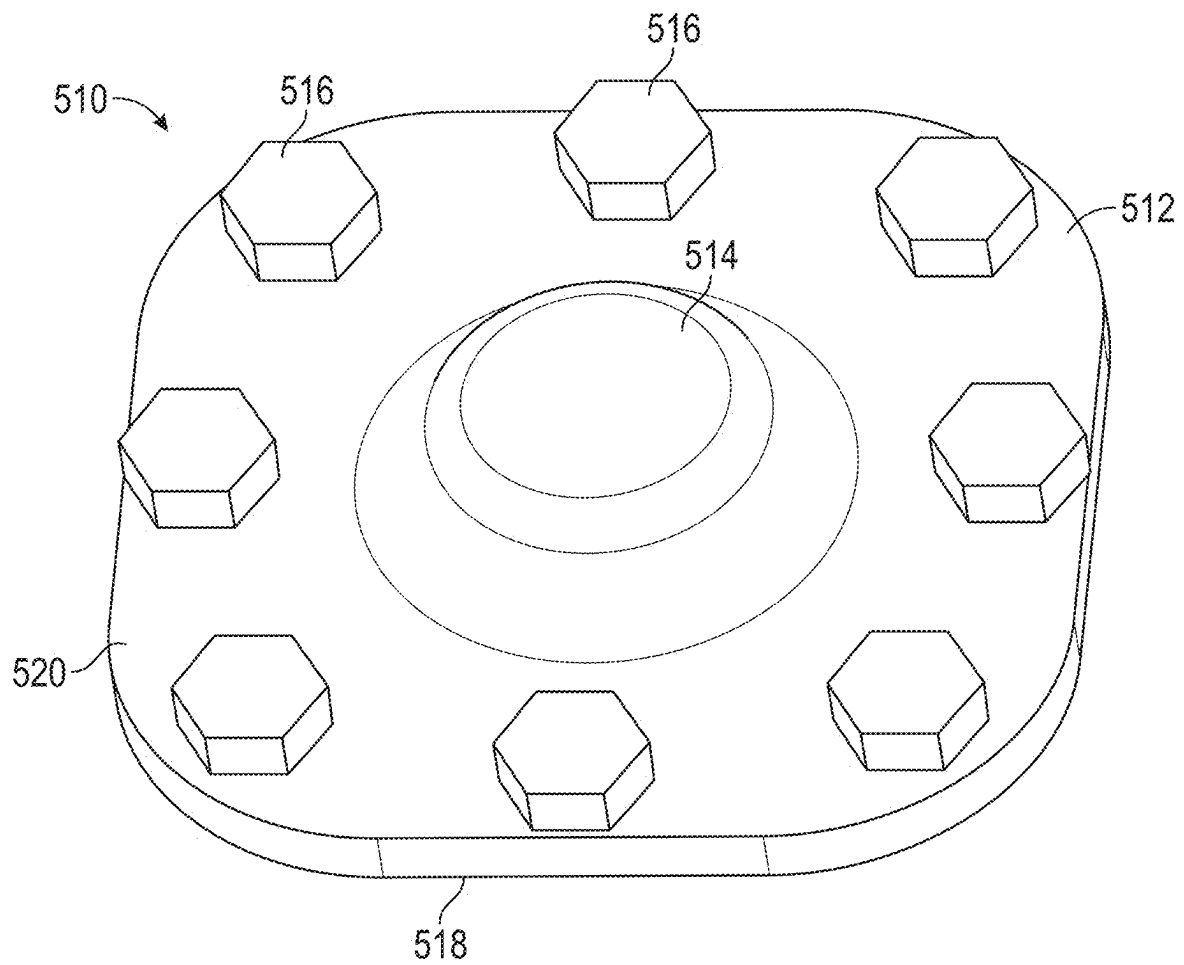


FIG. 6

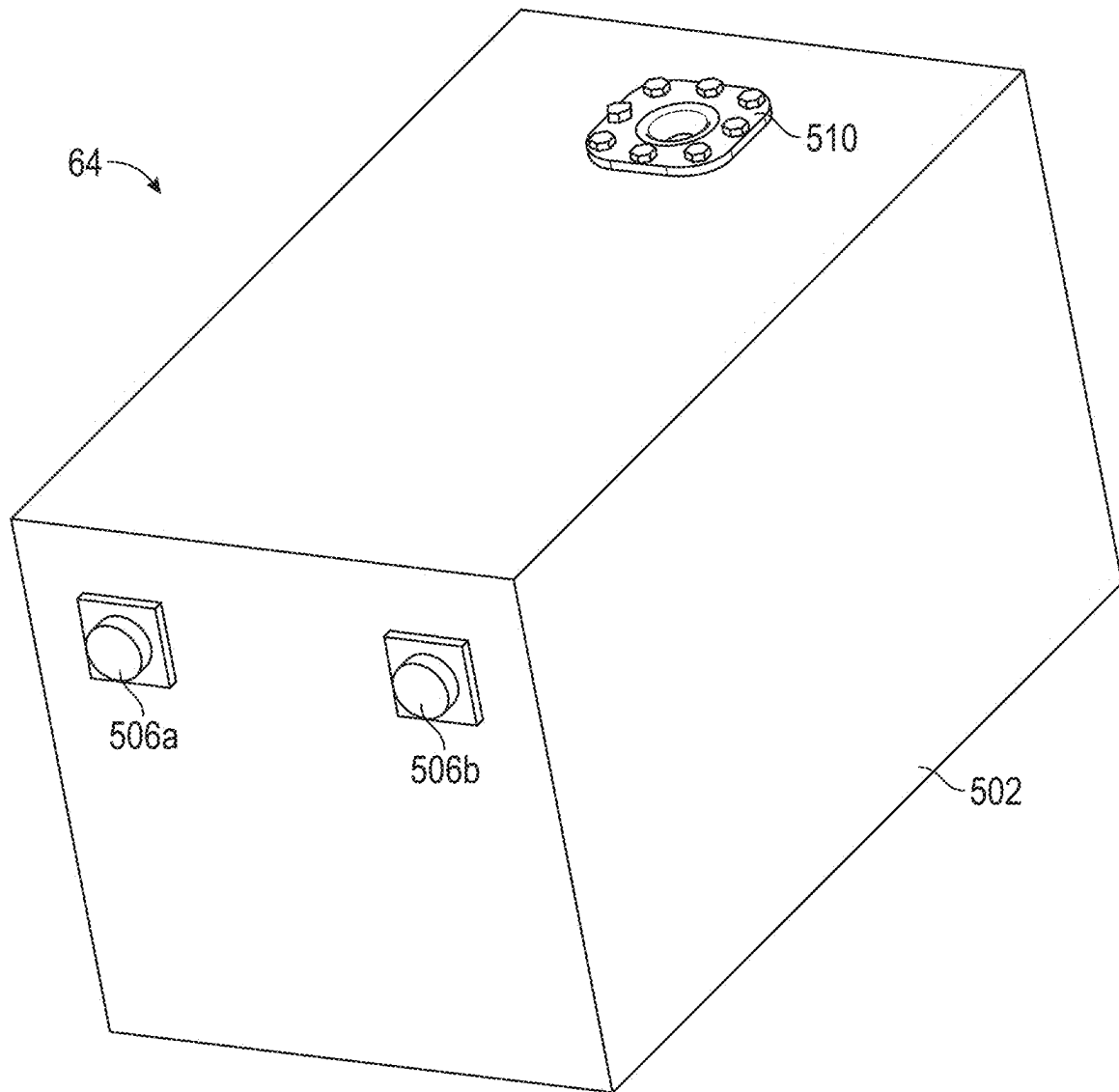


FIG. 7

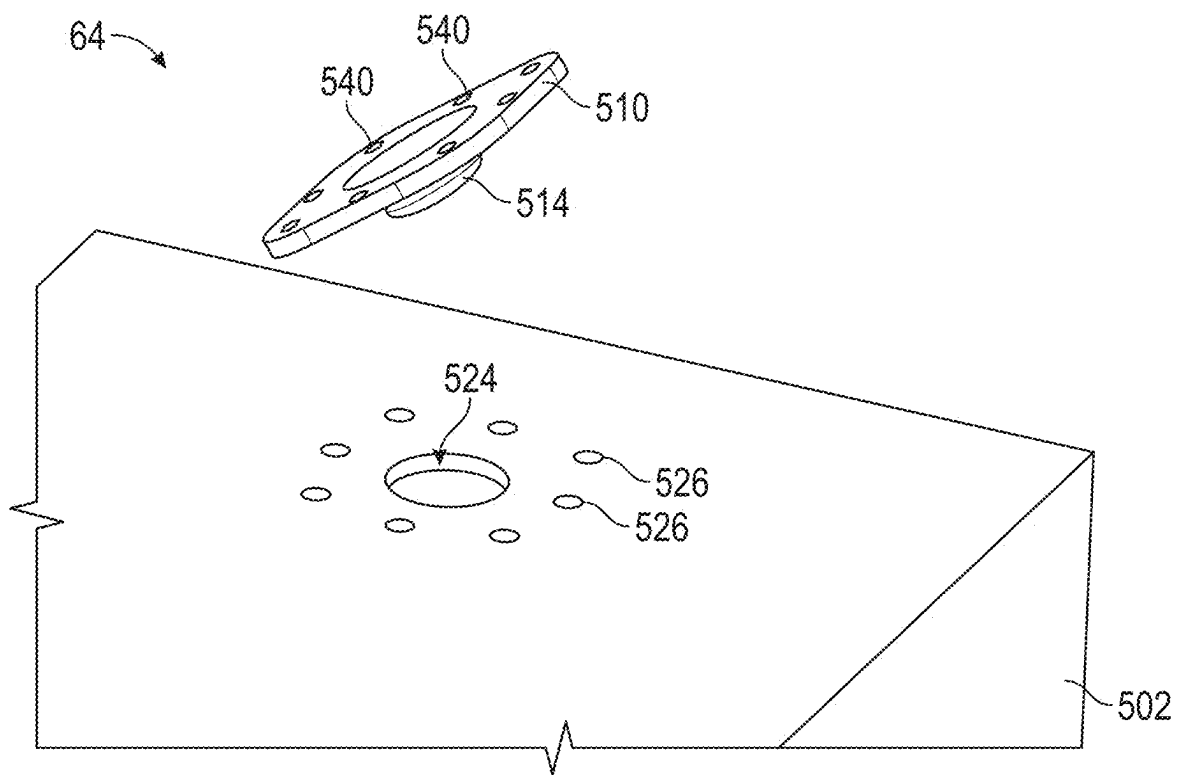


FIG. 8

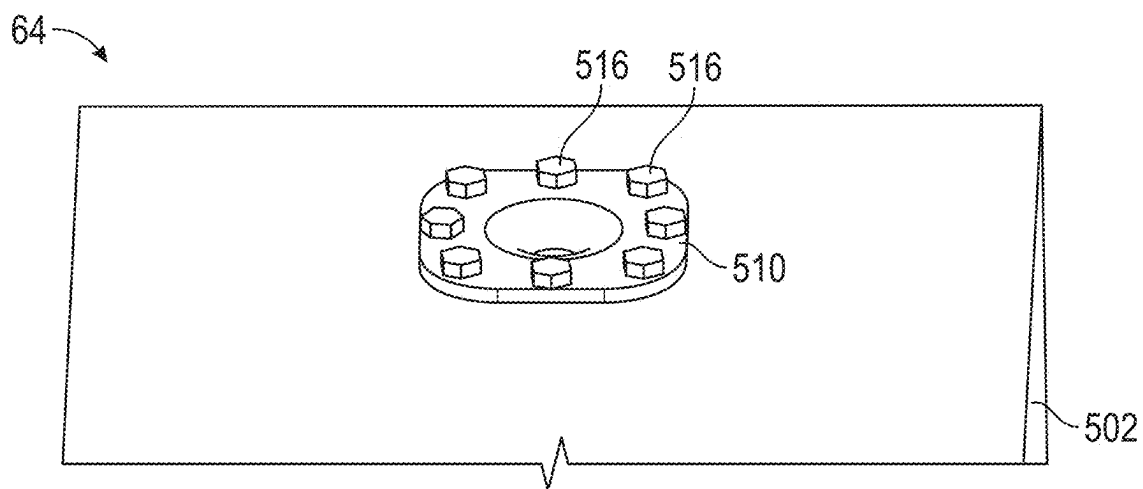


FIG. 9

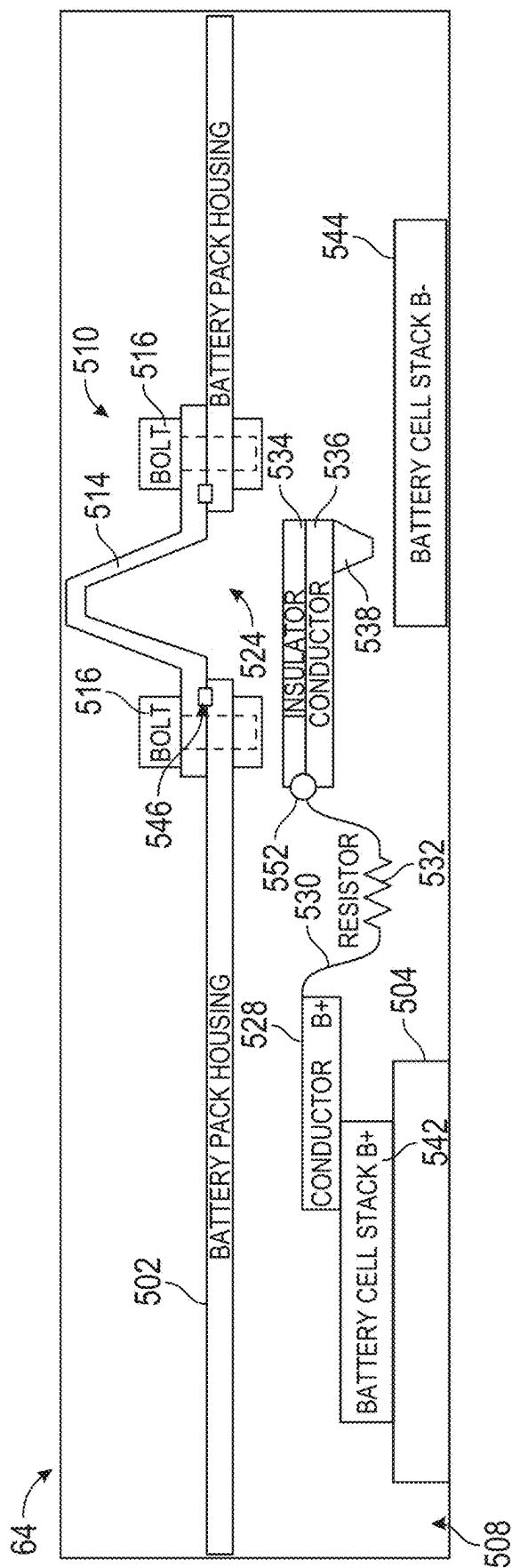


FIG. 10A

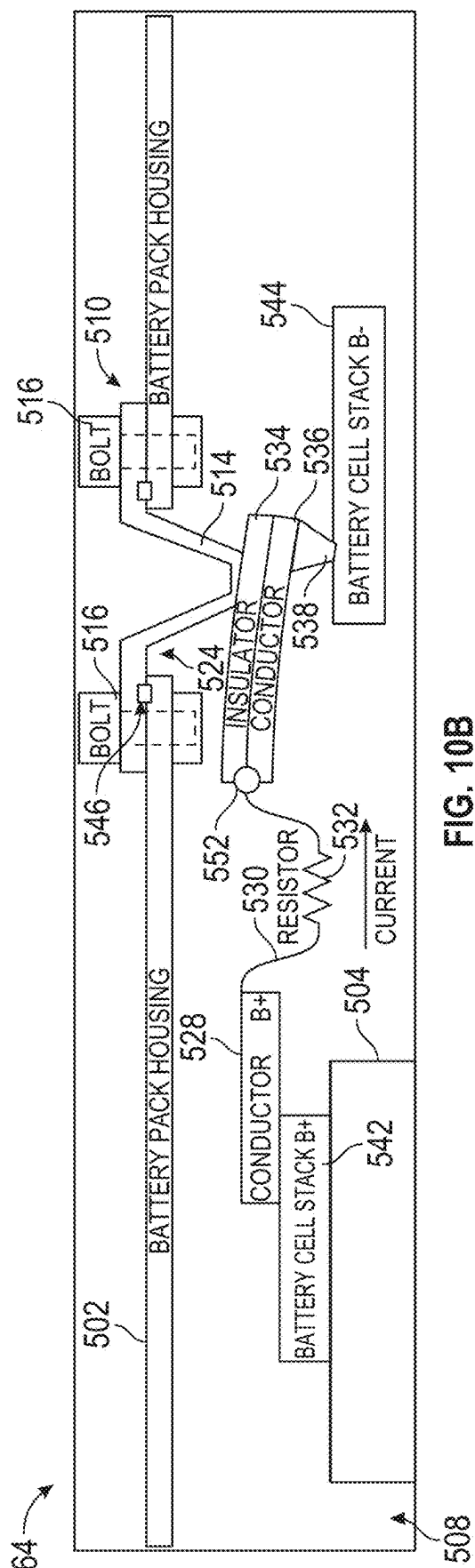


FIG. 10B

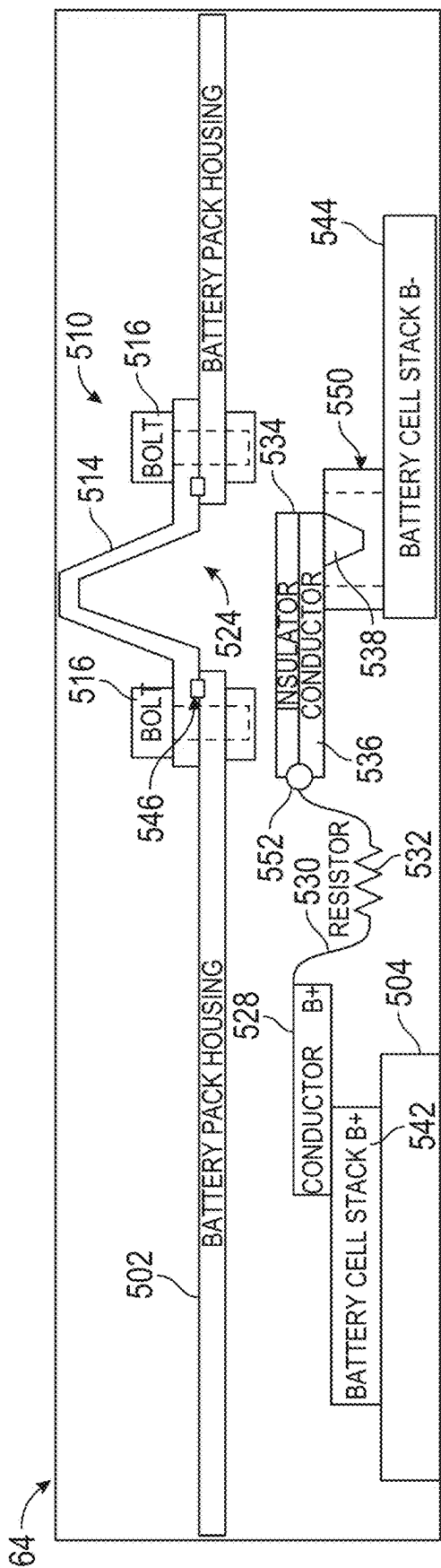


FIG. 11A

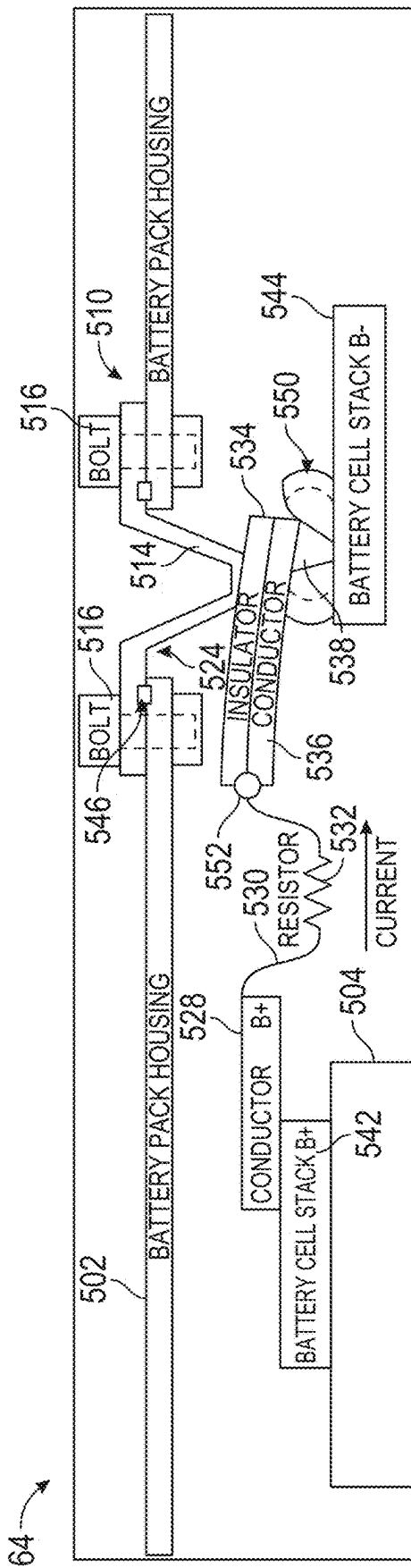


FIG. 11B

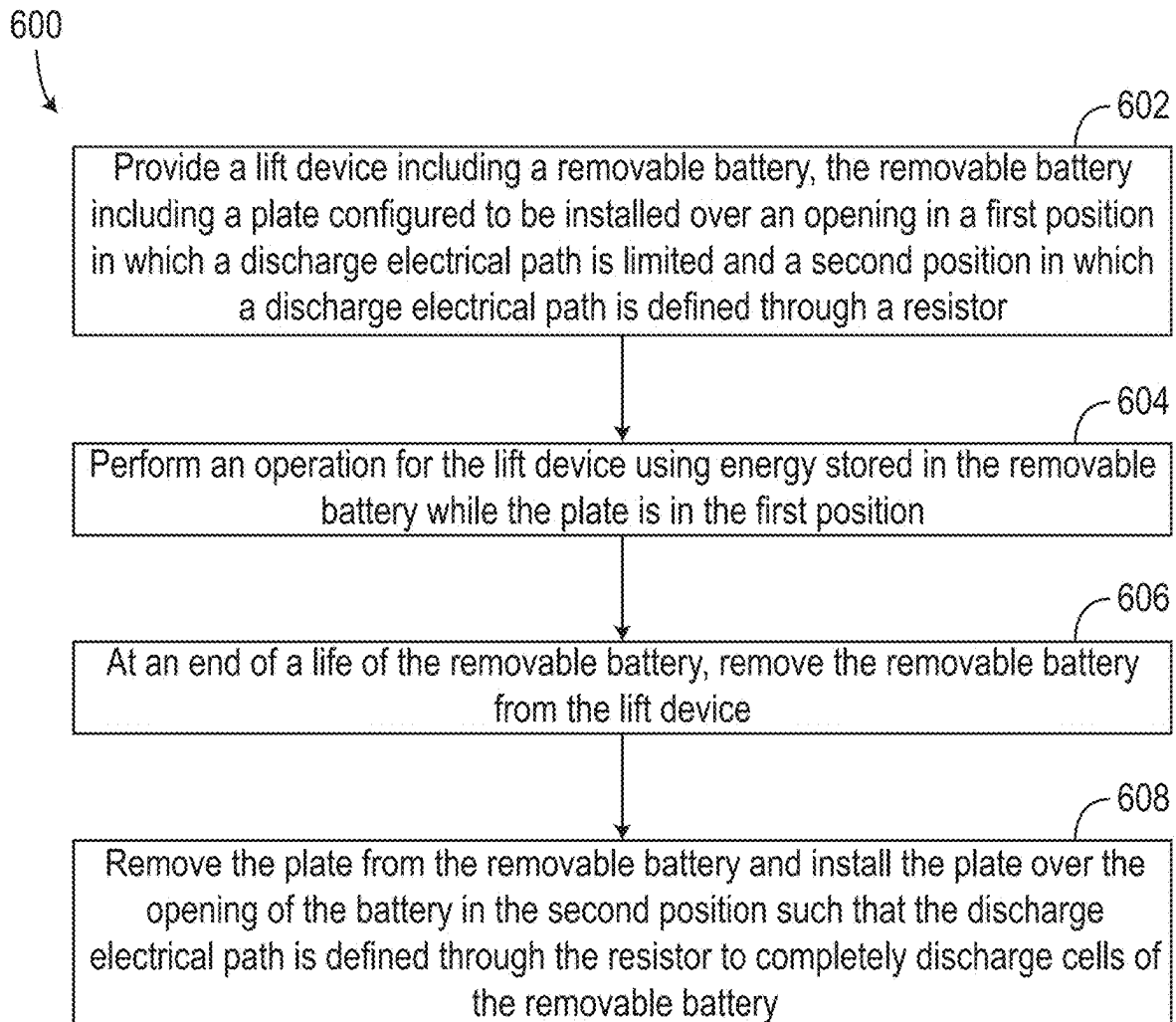


FIG. 12

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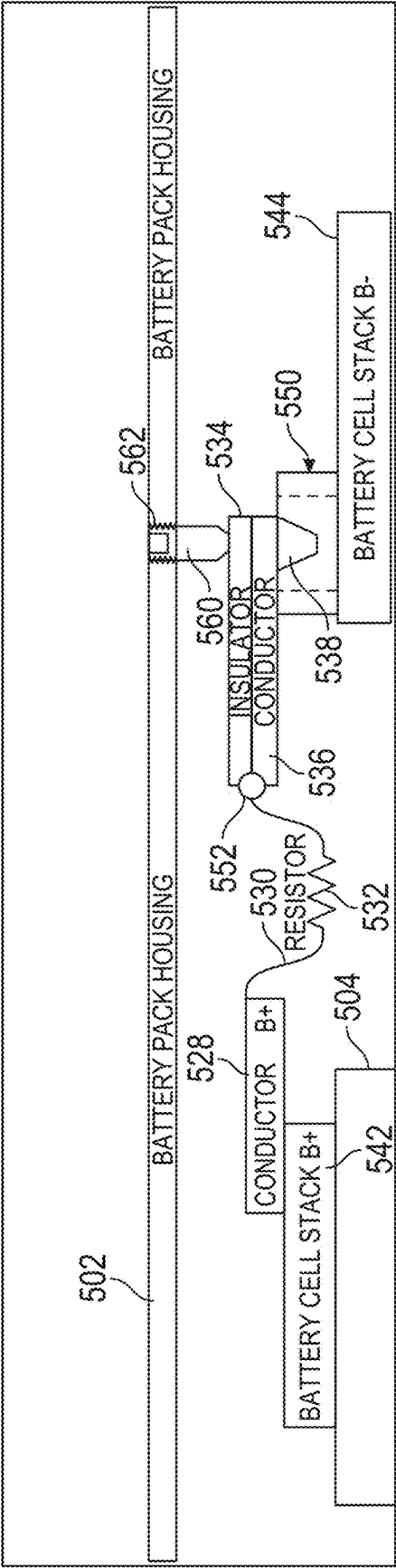


FIG. 13

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**LIFT DEVICE WITH INTERNALLY
DEPLETABLE BATTERY****CROSS-REFERENCE TO RELATED PATENT
APPLICATION**

This application claims the benefit of and priority to U.S. Provisional Application No. 63/578,601, filed Aug. 24, 2023, the entire disclosure of which is incorporated by reference herein.

BACKGROUND

The present disclosure relates to lift devices. More specifically, the present disclosure relates to energy storage systems for lift devices.

SUMMARY

One embodiment of the present disclosure is a battery pack for a lift device. The battery pack includes a housing, multiple battery cells, a resistor, a conductive element, and a member. The housing includes an opening. The battery cells are positioned within the housing. The resistor is electrically coupled with a positive terminal of the plurality of battery cells and is positioned within the housing. The conductive element is positioned within the housing and is configured to transition between an open state in which a discharge path is not defined between the positive terminal of the battery cells and a negative terminal or a ground, and a closed state in which the discharge path is defined between the positive terminal of the battery cells to the negative terminal or the ground through the resistor. The member is disposed at the opening and accessible from an exterior of the housing. The member is manually transitionable by a technician between a first state in which the conductive element is in the first state, and a second state in which the conductive element is driven into the second state such that the plurality of battery cells discharge remaining electrical energy via the discharge path.

In some embodiments, the member includes a plate having a protrusion in a center. In the first state, the member is fastened over the opening such that the protrusion is external to the housing. In the second state, the member is fastened over the opening such that the protrusion extends into the housing through the opening and drives the conductive element into the closed state.

In some embodiments, the member includes a screw configured to be received within the opening. The screw is configured to be accessed from the exterior of the housing by the technician such that the screw is driven to rotate to translate into the second state to bias the conductive element into the closed state.

In some embodiments, the conductive element includes a cantilever beam having a fixed end and a free end. The conductive element is configured to be driven by the member to bend such that a protrusion of the free end engages the negative terminal or the ground in the closed state.

In some embodiments, the battery pack includes an insulator disposed on a side of the conductive element opposite a side from which a protrusion extends. The member is configured to engage the insulator to transition the conductive element from the open state to the closed state.

In some embodiments, the battery pack includes a spacer disposed on a tip of the conductive element. The spacer is configured to align the tip of the conductive element and deform as the conductive element is driven by the member

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to the closed state. In some embodiments, conductive element includes a cantilever beam having a fixed end coupled with the housing and a free end configured to be driven to engage the negative terminal or the ground in the closed state.

Another embodiment of the present disclosure is a lift device. The lift device includes a lift assembly, and a battery pack. The lift assembly is configured to raise or lower. The battery pack is configured to provide electrical energy to the lift assembly. The battery pack includes a housing, battery cells, a resistor, a conductive element, and a member. The housing includes an opening. The battery cells are positioned within the housing. The resistor is electrically coupled with a positive terminal of the battery cells and is positioned within the housing. The conductive element is positioned within the housing and is configured to transition between an open state in which a discharge path is not defined between the positive terminal of the battery cells and a negative terminal or a ground, and a closed state in which the discharge path is defined between the positive terminal of the battery cells to the negative terminal or the ground through the resistor. The member is disposed at the opening and is accessible from an exterior of the housing. The member is manually transitionable by a technician between a first state in which the conductive element is in the first state, and a second state in which the conductive element is driven into the second state such that the battery cells discharge remaining electrical energy via the discharge path.

In some embodiments, the member includes a plate having a protrusion in a center. In the first state the member is fastened over the opening such that the protrusion is external to the housing. In the second state, the member is fastened over the opening such that the protrusion extends into the housing through the opening and drives the conductive element into the closed state.

In some embodiments, the member includes a screw configured to be received within the opening. The screw is configured to be accessed from the exterior of the housing by the technician such that the screw is driven to rotate to translate into the second state to bias the conductive element into the closed state.

In some embodiments, the conductive element includes a cantilever beam having a fixed end and a free end. The conductive element is configured to be driven by the member to bend such that a protrusion of the free end engages the negative terminal or the ground in the closed state.

In some embodiments, the battery pack comprises an insulator disposed on a side of the conductive element opposite a side from which a protrusion extends. The member is configured to engage the insulator to transition the conductive element from the open state to the closed state.

In some embodiments, the battery pack further includes a spacer disposed on a tip of the conductive element. The spacer is configured to align the tip of the conductive element and deform as the conductive element is driven by the member to the closed state.

In some embodiments, the conductive element includes a cantilever beam having a fixed end coupled with the housing and a free end configured to be driven to engage the negative terminal or the ground in the closed state. In some embodiments, the lift device is a fully electric boom.

Another embodiment of the present disclosure is a method of completely discharging a battery. The method includes providing a device including a removable battery. The removable battery includes a member configured to be transitioned between a first state and a second state. The

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method also includes performing an operation with the device using energy provided by the removable battery. The method also includes removing the removable battery from the device. The method also includes transitioning the member from the first state into the second state such that a discharge electrical path is defined across terminals of cells of the removable battery through a resistor to completely discharge the cells of the removable battery.

In some embodiments, transitioning the member from the first state to the second state includes removing the member from a side of a housing of the removable battery. The member includes a protrusion on one side with the protrusion oriented in an outwards direction when the member is in the first state. The method also includes re-orienting and reinstalling the member in the second state on the side of the housing such that the protrusion extends through an opening in the housing of the removable battery and biases a conductive member to contact a negative terminal or ground of the removable battery such that energy is depleted from cells of the removable battery through a resistor.

In some embodiments, transitioning the member from the first state to the second state includes screwing the member into a side of a housing of the removable battery such that the member protrudes further into the removable battery and drives a conductive member to contact a negative terminal or ground of the removable battery such that energy is depleted from cells of the removable battery through a resistor. In some embodiments, transitioning the member from the first state to the second state drives a conductive member having the form of a cantilever beam into engagement with a ground or a negative terminal of the removable battery. The conductive member is electrically coupled with a positive terminal of the removable battery through a resistor. In some embodiments, transitioning the member from the first state to the second state includes removing a fastener that couples the member with a side of a housing of the removable battery, re-orienting the member, and re-installing the fastener.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a lift device, according to some embodiments.

FIG. 2 is a perspective view of a base of the lift device of FIG. 1, according to some embodiments.

FIG. 3 is a perspective view of an axle assembly of the lift device of FIG. 1, according to some embodiments.

FIG. 4 is a perspective view of a platform assembly of the lift device of FIG. 1, according to some embodiments.

FIG. 5 is a perspective view of a battery of the lift device of FIG. 1, according to some embodiments.

FIG. 6 is a perspective view of a member of the battery of FIG. 5 including a protrusion on one side, according to some embodiments.

FIG. 7 is a perspective view of the battery of FIG. 5, according to some embodiments.

FIG. 8 is a perspective view of a portion of the top of the battery of FIG. 5 with the member of FIG. 7 removed, according to some embodiments.

FIG. 9 is a perspective view of a portion of the top of the battery of FIG. 5 with the member of FIG. 7 installed in a second position, according to some embodiments.

FIG. 10A is a sectional view of an interior of the battery of FIG. 5 including a discharge mechanism, with the member of FIG. 7 in the first position, according to some embodiments.

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FIG. 10B is a sectional view of the interior of the battery of FIG. 5 including the discharge mechanism, with the member of FIG. 7 in the second position, according to some embodiments.

FIG. 11A is a sectional view of the interior of the battery of FIG. 5 including a discharge mechanism that includes a spacer, with the member of FIG. 7 in the first position, according to some embodiments.

FIG. 11B is a sectional view of the interior of the battery of FIG. 5 including the discharge mechanism that includes the spacer, with the member of FIG. 7 in the second position, according to some embodiments.

FIG. 12 is a flow diagram of a process for using and discharging a battery of electrical equipment, according to some embodiments.

FIG. 13 is a sectional view of the interior of the battery of FIG. 5 including another discharge mechanism that includes a screw to initiate discharge of the battery, according to some embodiments.

DETAILED DESCRIPTION

Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

Overview

Referring generally to the FIGURES, a lift device includes a battery having multiple cells positioned within a housing that are configured to receive, store, and discharge electrical energy for one or more electrical loads of the lift device (e.g., an electric motor, an electric linear actuator, lighting devices, controllers, etc.). The battery includes a mechanism that is configured to be manually activated in order to discharge the cells of the battery (e.g., to convert the electrical energy into heat). In some embodiments, when the battery is about to be transported (e.g., due to defects, servicing, or end-of-life conditions), the mechanism can be manually activated in order to ensure that the cells of the battery are completely or substantially depleted of energy before handling and shipping of the battery.

Lift Device

Referring to FIG. 1, a lifting apparatus, lift device, or mobile elevating work platform (MEWP) (e.g., a telehandler, an electric boom lift, a towable boom lift, a lift device, a fully electric boom lift, etc.), shown as lift device 10 includes a base assembly 12 (e.g., a base, a support assembly, a drivable support assembly, a support structure, a chassis, etc.), a platform assembly 16 (e.g., a platform, a terrace, etc.), and a lift assembly 14 (e.g., a boom, a boom lift assembly, a lifting apparatus, an articulated arm, a scissors lift, etc.). The lift device 10 includes a front end (e.g., a forward-facing end, a front portion, a front, etc.), shown as front 62, and a rear end (e.g., a rearward facing end, a back portion, a back, a rear, etc.) shown as rear 60. The lift assembly 14 is configured to elevate the platform assembly 16 in an upward direction 46 (e.g., an upward vertical direction) relative to the base assembly 12. The lift assembly 14 is also configured to translate the platform assembly 16 in a downward direction 48 (e.g., a downward vertical direction). The lift assembly 14 is also configured to translate the platform assembly 16 in either a forward

direction **50** (e.g., a forward longitudinal direction) or a rearward direction **51** (e.g., a rearward longitudinal direction). The lift assembly **14** generally facilitates performing a lifting function to raise and lower the platform assembly **16**, as well as movement of the platform assembly **16** in various directions.

The base assembly **12** defines a longitudinal axis **78** and a lateral axis **80**. The longitudinal axis **78** defines the forward direction **50** of lift device **10** and the rearward direction **51**. The lift device **10** is configured to translate in the forward direction **50** and to translate backwards in the rearward direction **51**. The base assembly **12** includes one or more wheels, tires, wheel assemblies, tractive elements, rotary elements, treads, etc., shown as tractive elements **82**. The tractive elements **82** are configured to rotate to drive (e.g., propel, translate, steer, move, etc.) the lift device **10**. The tractive elements **82** can each include an electric motor **52** (e.g., electric wheel motors) configured to drive the tractive elements **82** (e.g., to rotate tractive elements **82** to facilitate motion of the lift device **10**). In other embodiments, the tractive elements **82** are configured to receive power (e.g., rotational mechanical energy) from electric motors **52** or through a drive train (e.g., a combination of any number and configuration of a shaft, an axle, a gear reduction, a gear train, a transmission, etc.). In some embodiments, one or more tractive elements **82** are driven by a prime mover **41** (e.g., electric motor, internal combustion engine, etc.) through a transmission. In some embodiments, a hydraulic system (e.g., one or more pumps, hydraulic motors, conduits, valves, etc.) transfers power (e.g., mechanical energy) from one or more electric motors **52** and/or the prime mover **41** to the tractive elements **82**. The tractive elements **82** and electric motors **52** (or prime mover **41**) can facilitate a driving and/or steering function of the lift device **10**. In some embodiments, the electric motors **52** are optional, and the tractive elements **82** are powered or driven by an internal combustion engine.

With additional reference to FIG. 4, the platform assembly **16** is shown in further detail. The platform assembly **16** is configured to provide a work area for an operator of the lift device **10** to stand/rest upon. The platform assembly **16** can be pivotally coupled to an upper end of the lift assembly **14**. The lift device **10** is configured to facilitate the operator accessing various elevated areas (e.g., lights, platforms, the sides of buildings, building scaffolding, trees, power lines, etc.). The lift device **10** may use various electrically-powered motors and electrically-powered linear actuators or hydraulic cylinders to facilitate elevation and/or horizontal movement (e.g., lateral movement, longitudinal movement) of the platform assembly **16** (e.g., relative to the base assembly **12**, or to a ground surface that the base assembly **12** rests upon). In some embodiments, the lift device **10** uses internal combustion engines, hydraulics, a hydraulic system, pneumatic cylinders, etc.

The platform assembly **16** includes a base member, a base portion, a platform, a standing surface, a shelf, a work platform, a floor, a deck, etc., shown as a deck **18**. The deck **18** provides a space (e.g., a floor surface) for a worker to stand upon as the platform assembly **16** is raised and lowered.

The platform assembly **16** includes a railing assembly including various members, beams, bars, guard rails, rails, railings, etc., shown as rails **22**. The rails **22** extend along substantially an entire perimeter of the deck **18**. The rails **22** provide one or more members for the operator of the lift device **10** to grasp while using the lift device **10** (e.g., to grasp while operating the lift device **10** to elevate the

platform assembly **16**). The rails **22** can include members that are substantially horizontal to the deck **18**. The rails **22** can also include vertical structural members that couple with the substantially horizontal members. The vertical structural members can extend upwards from the deck **18**.

The platform assembly **16** can include a human machine interface (HMI) (e.g., a user interface, an operator interface, etc.), shown as the user interface **20**. The user interface **20** is configured to receive user inputs from the operator at or upon the platform assembly **16** to facilitate operation of the lift device **10**. The user interface **20** can include any number of buttons, levers, switches, keys, etc., or any other user input device configured to receive a user input to operate the lift device **10**. The user interface **20** may also provide information to the user (e.g., through one or more displays, lights, speakers, haptic feedback devices, etc.). The user interface **20** can be supported by one or more of the rails **22**.

Referring to FIG. 1, the platform assembly **16** includes a frame **24** (e.g., structural members, support beams, a body, a structure, etc.) that extends at least partially below the deck **18**. The frame **24** can be integrally formed with the deck **18**. The frame **24** is configured to provide structural support for the deck **18** of the platform assembly **16**. The frame **24** can include any number of structural members (e.g., beams, bars, I-beams, etc.) to support the deck **18**. The frame **24** couples the platform assembly **16** with the lift assembly **14**. The frame **24** may be rotatably or pivotally coupled with the lift assembly **14** to facilitate rotation of the platform assembly **16** about an axis **28** (e.g., a vertical axis). The frame **24** can also rotatably/pivotally couple with the lift assembly **14** such that the frame **24** and the platform assembly **16** can pivot about an axis **25** (e.g., a horizontal axis).

The lift assembly **14** includes one or more beams, articulated arms, bars, booms, arms, support members, boom sections, cantilever beams, etc., shown as lift arms **32a**, **32b**, and **32c**. The lift arms are hingedly or rotatably coupled with each other at their ends. The lift arms can be hingedly or rotatably coupled to facilitate articulation of the lift assembly **14** and raising/lowering and/or horizontal movement of the platform assembly **16**. The lift device **10** includes a lower lift arm **32a**, a central or medial lift arm **32b**, and an upper lift arm **32c**. The lower lift arm **32a** is configured to hingedly or rotatably couple at one end with the base assembly **12** to facilitate lifting (e.g., elevation) of the platform assembly **16**. The lower lift arm **32a** is configured to hingedly or rotatably couple at an opposite end with the medial lift arm **32b**. Likewise, the medial lift arm **32b** is configured to hingedly or rotatably couple with the upper lift arm **32c**. The upper lift arm **32c** can be configured to hingedly interface/couple and/or telescope with an intermediate lift arm **32d**. The upper lift arm **32c** can be referred to as “the jib” of the lift device **10**. The intermediate lift arm **32d** may extend into an inner volume of the upper lift arm **32c** and extend and/or retract. The lower lift arm **32a** and the medial lift arm **32b** may be referred to as “the boom” of the overall lift device **10** assembly. The intermediate lift arm **32d** can be configured to couple (e.g., rotatably, hingedly, etc.), with the platform assembly **16** to facilitate levelling of the platform assembly **16**.

The lift arms **32** are driven to hinge or rotate relative to each other by actuators **34a**, **34b**, **34c**, and **34d** (e.g., electric linear actuators, linear electric arm actuators, hydraulic cylinders, etc.). The actuators **34a**, **34b**, **34c**, and **34d** can be mounted between adjacent lift arms to drive adjacent lift arms to hinge or pivot (e.g., rotate some angular amount) relative to each other about pivot points **84**. The actuators **34a**, **34b**, **34c**, and **34d** can be mounted between adjacent lift

arms using any of a foot bracket, a flange bracket, a clevis bracket, a trunnion bracket, etc. The actuators **34a**, **34b**, **34c**, and **34d** may be configured to extend or retract (e.g., increase in overall length, or decrease in overall length) to facilitate pivoting adjacent lift arms to pivot/hinge relative to each other, thereby articulating the lift arms and raising or lowering the platform assembly **16**.

The actuators **34a**, **34b**, **34c**, and **34d** can be configured to extend (e.g., increase in length) to increase a value of an angle formed between adjacent lift arms **32**. The angle can be defined between centerlines of adjacent lift arms **32** (e.g., centerlines that extend substantially through a center of the lift arms **32**). For example, the actuator **34a** is configured to extend/retract to increase/decrease the angle **75a** defined between a centerline of the lower lift arm **32a** and the longitudinal axis **78** (angle **75a** can also be defined between the centerline of the lower lift arm **32a** and a plane defined by the longitudinal axis **78** and lateral axis **80**) and facilitate lifting of the platform assembly **16** (e.g., moving the platform assembly **16** at least partially along the upward direction **46**). Likewise, the actuator **34b** can be configured to retract to decrease the angle **75a** to facilitate lowering of the platform assembly **16** (e.g., moving the platform assembly **16** at least partially along the downward direction **48**). Similarly, the actuator **34b** is configured to extend to increase the angle **75b** defined between centerlines of the lower lift arm **32a** and the medial lift arm **32b** and facilitate elevating of the platform assembly **16**. Similarly, the actuator **34b** is configured to retract to decrease the angle **75b** to facilitate lowering of the platform assembly **16**. The electric actuator **34c** is similarly configured to extend/retract to increase/decrease the angle **75c**, respectively, to raise/lower the platform assembly **16**. The actuators **34** may be hydraulic actuators, electric actuators, pneumatic actuators, etc.

The actuators **34a**, **34b**, **34c**, and **34d** can be mounted (e.g., rotatably coupled, pivotally coupled, etc.) to adjacent lift arms at mounts **40** (e.g., mounting members, mounting portions, attachment members, attachment portions, etc.). The mounts **40** can be positioned at any position along a length of each lift arm. For example, the mounts **40** can be positioned at a midpoint of each lift arm, and a lower end of each lift arm.

The intermediate lift arm **32d** and the frame **24** are configured to pivotally interface/couple at a platform rotator **30** (e.g., a rotary actuator, a rotational electric actuator, a gear box, etc.). The platform rotator **30** facilitates rotation of the platform assembly **16** about the axis **28** relative to the intermediate lift arm **32d**. In some embodiments, the platform rotator **30** is positioned between the frame **24** and the upper lift arm **32c** and facilitates pivoting of the platform assembly **16** relative to the upper lift arm **32c**. The axis **28** extends through a central pivot point of the platform rotator **30**. The intermediate lift arm **32d** can also be configured to articulate or bend such that a distal portion of the intermediate lift arm **32d** pivots/rotates about the axis **25**. The intermediate lift arm **32d** can be driven to rotate/pivot about axis **25** by extension and retraction of the actuator **34d**.

The intermediate lift arm **32d** is also configured to extend/retract (e.g., telescope) along the upper lift arm **32c**. In some embodiments, the lift assembly **14** includes a linear actuator (e.g., a hydraulic cylinder, an electric linear actuator, etc.), shown as extension actuator **35**, that controls extension and retraction of the intermediate lift arm **32d** relative to the upper lift arm **32c**. In other embodiments, one more of the other arms of the lift assembly **14** include multiple telescoping sections that are configured to extend/retract relative to one another.

The platform assembly **16** is configured to be driven to pivot about the axis **28** (e.g., rotate about axis **28** in either a clockwise or a counter-clockwise direction) by an electric or hydraulic motor **26** (e.g., a rotary electric actuator, a stepper motor, a platform rotator, a platform electric motor, an electric platform rotator motor, etc.). The motor **26** (e.g., the pivot motor **26**) can be configured to drive the frame **24** to pivot about the axis **28** relative to the upper lift arm **32c** (or relative to the intermediate lift arm **32d**). The motor **26** can be configured to drive a gear train to pivot the platform assembly **16** about the axis **28**.

Referring to FIGS. **1** and **2**, the lift assembly **14** is configured to pivotally or rotatably couple with the base assembly **12**. The base assembly **12** includes a rotatable base member, a rotatable platform member, a fully electric turntable, etc., shown as a turntable **70**. The lift assembly **14** is configured to rotatably/pivotally couple with the base assembly **12**. The turntable **70** is rotatably coupled with a base, frame, structural support member, carriage, etc., of base assembly **12**, shown as base **36**. The turntable **70** is configured to rotate or pivot relative to the base **36**. The turntable **70** can pivot/rotate about the central axis **42** relative to base **36**, about a slew bearing **71** (e.g., the slew bearing **71** pivotally couples the turntable **70** to the base **36**). The turntable **70** facilitates accessing various elevated and angularly offset locations at the platform assembly **16**. The turntable **70** is configured to be driven to rotate or pivot relative to base **36** and about the slew bearing **71** by an electric motor, an electric turntable motor, an electric rotary actuator, a hydraulic motor, etc., shown as the turntable motor **44**. The turntable motor **44** can be configured to drive a geared outer surface **73** of the slew bearing **71** that is rotatably coupled to the base **36** about the slew bearing **71** to rotate the turntable **70** relative to the base **36**. The lower lift arm **32a** is pivotally coupled with the turntable **70** (or with a turntable member **72** of the turntable **70**) such that the lift assembly **14** and the platform assembly **16** rotate as the turntable **70** rotates about the central axis **42**. In some embodiments, the turntable **70** is configured to rotate a complete 360 degrees about the central axis **42** relative to the base **36**. In other embodiments, the turntable **70** is configured to rotate an angular amount less than 360 degrees about the central axis **42** relative to the base **36** (e.g., 270 degrees, 120 degrees, etc.).

The base assembly **12** includes one or more energy storage devices or power sources (e.g., capacitors, batteries, Lithium-Ion batteries, Nickel Cadmium batteries, fuel tanks, etc.), shown as batteries **64** (e.g., battery packs). The batteries **64** are configured to store energy in a form (e.g., in the form of chemical energy) that can be converted into electrical energy for the various electric motors and actuators of the lift device **10**. The batteries **64** can be stored within the base **36**. The lift device **10** includes a controller **38** that is configured to operate any of the motors, actuators, etc., of the lift device **10**. The controller **38** can be configured to receive sensory input information from various sensors of the lift device **10**, user inputs from the user interface **20** (or any other user input device such as a key-start or a push-button start), etc. The controller **38** can be configured to generate control signals for the various motors, actuators, etc., of the lift device **10** to operate any of the motors, actuators, electrically powered movers, etc., of the lift device **10**. The batteries **64** are configured to power any of the motors, sensors, actuators, electric linear actuators, electrical devices, electrical movers, stepper motors, etc., of the lift device **10**. The base assembly **12** can include a power circuit including any necessary transformers, resistors, tran-

sistors, thermistors, capacitors, etc., to provide appropriate power (e.g., electrical energy with appropriate current and/or appropriate voltage) to any of the motors, electric actuators, sensors, electrical devices, etc., of the lift device 10.

The batteries 64 are configured to deliver power to the motors 52 to drive the tractive elements 82. A rear set of tractive elements 82 can be configured to pivot to steer the lift device 10. In other embodiments, a front set of tractive elements 82 are configured to pivot to steer the lift device 10. In still other embodiments, both the front and the rear set of tractive elements 82 are configured to pivot (e.g., independently) to steer the lift device 10. In some examples, the base assembly 12 includes a steering system 150. The steering system 150 is configured to drive tractive elements 82 to pivot for a turn of the lift device 10. The steering system 150 can be configured to pivot the tractive elements 82 in pairs (e.g., to pivot a front pair of tractive elements 82), or can be configured to pivot tractive elements 82 independently (e.g., four-wheel steering for tight-turns).

It should be understood that while the lift device 10 as described herein is described with reference to batteries, electric motors, etc., the lift device 10 can be powered (e.g., for transportation and/or lifting the platform assembly 16) using one or more internal combustion engines, electric motors or actuators, hydraulic motors or actuators, pneumatic actuators, or any combination thereof.

In some embodiments, the base assembly 12 also includes a user interface 21 (e.g., a HMI, a user interface, a user input device, a display screen, etc.). In some embodiments, the user interface 21 is coupled to the base 36. In other embodiments, the user interface 21 is positioned on the turntable 70. The user interface 21 can be positioned on any side or surface of the base assembly 12 (e.g., on the front 62 of the base 36, on the rear 60 of the base 36, etc.).

Referring now to FIGS. 2 and 3, the base assembly 12 includes a longitudinally extending frame member 54 (e.g., a rigid member, a structural support member, an axle, a base, a frame, a carriage, a chassis, etc.). The longitudinally extending frame member 54 provides structural support for the turntable 70 as well as the tractive elements 82. The longitudinally extending frame member 54 is pivotally coupled with lateral frame members 110 (e.g., axles, frame members, beams, bars, etc.) at opposite longitudinal ends of the longitudinally extending frame member 54. For example, the lateral frame members 110 may be pivotally coupled with the longitudinally extending frame member 54 at a front end and a rear end of the longitudinally extending frame member 54. The lateral frame members 110 can each be configured to pivot about a pivot joint 58 (e.g., about a longitudinal axis). The pivot joint 58 can include a pin and a receiving portion (e.g., a bore, an aperture, etc.). The pin of the pivot joint 58 is coupled to one of the lateral frame members 110 (e.g., a front lateral frame member 110 or a rear lateral frame member 110) or the longitudinally extending frame member 54 and the receiving portion is coupled to the other of the longitudinally extending frame member 54 and the lateral frame member 110. For example, the pin may be coupled with longitudinally extending frame member 54 and the receiving portion can be coupled with one of the lateral frame members 110 (e.g., integrally formed with the front lateral frame member 110).

In some embodiments, the longitudinally extending frame member 54 and the lateral frame members 110 are integrally formed or coupled (e.g., fastened, welded, riveted, etc.) to define the base 36. In still other embodiments, the base 36 is integrally formed with the longitudinally extending frame member 54 and/or the lateral frame members 110. In still

other embodiments, the base 36 is coupled with the longitudinally extending frame member 54 and/or the lateral frame members 110.

The base assembly 12 includes one or more axle actuators 56 (e.g., electric linear actuators, electric axle actuators, electric levelling actuators, hydraulic cylinders, etc.). The axle actuators 56 can be linear actuators configured to receive power from the batteries 64, for example. The axle actuators 56 can be configured to extend or retract to contact a top surface of a corresponding one of the lateral frame members 110. When the axle actuators 56 extend, an end of a rod of the levelling actuators can contact the surface of lateral frame member 110 and prevent relative rotation between lateral frame member 110 and longitudinally extending frame member 54. In this way, the relative rotation/pivoting between the lateral frame member 110 and the longitudinally extending frame member 54 can be locked (e.g., to prevent rolling of the longitudinally extending frame member 54 relative to the lateral frame members 110 during operation of the lift assembly 14). The axle actuators 56 can receive power from the batteries 64, which can allow the axle actuators 56 to extend or retract. The axle actuators 56 receive control signals from controller 38.

Manually Dischargeable Batteries

Referring to FIGS. 5-11, the batteries 64 may each include a housing 502 (e.g., a shell, a structure, a modular unit, walls, sidewalls, an enclosure, a container, a capsule, a cover, a case, a covering, casing, a hull, etc.), that defines an inner volume 508 (e.g., a space, a void, an area, etc.) within which one or more cells 504 (e.g., battery cells, energy storage cells, alkaline cells, lithium ion cells, etc.) are positioned. The cells 504 may be electrically coupled (e.g., via wiring) with one or more terminals 506 (e.g., a positive terminal 506a and a negative terminal 506b) that are accessible from an exterior of the housing 502 in order to discharge electrical energy for devices (e.g., actuators, motors, etc.) that consume electrical energy in order to operate (e.g., electrical components of the lift device 10). In some embodiments, the batteries 64 each include a connector or power connector such that the cells 504 of the batteries 64 can be quickly electrically coupled or de-coupled with other batteries 64 of the lift device 10 or with an energy distribution and discharge system of the lift device 10. The batteries 64 may be high-voltage or low-voltage batteries 64 configured to supply electrical energy for one or more high-voltage or low-voltage electrical devices of the lift device 10.

Referring particularly to FIGS. 5-9, the battery 64 may include a cap (e.g., a cover, a stopper, a lid, a plug, etc.), shown as manual discharge member 510. The manual discharge member 510 may be a structural member that functions to both initiate a manual discharge and as a housing member that seals with the housing 502 and forms or defines a portion of the housing 502. In some embodiments, the manual discharge member 510 is a removable, adjustable, repositionable, or re-orientable portion or member of the housing 502. The manual discharge member 510 is configured to removably couple with the housing 502 in a first position (e.g., a first state) or orientation where the manual discharge member 510 functions as a portion of the housing 502, and a second position or orientation where the manual discharge member 510 functions to initiate a complete discharge of the cells 504 (e.g., a closed state or position, a second state). In some embodiments, the manual discharge member 510 is positioned over an opening 524 (e.g., a hole, a window, an aperture, a bore, a space, etc.) and is configured to be fastened to the housing 502 via one or more fasteners 516 and openings 526 that are formed or

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define in the housing 502 and disposed in an array or pattern surrounding the opening 524. In some embodiments, the manual discharge member 510 includes corresponding openings 540 that are configured to receive the fasteners 516 therethrough such that the manual discharge member 510 can be fastened to the housing 502 over the opening 524.

Referring particularly to FIGS. 6 and 8, the manual discharge member 510 may have the form of a plate or planar member, shown as plate 512 having a first side 518, and a second side 520 (e.g., a first and second surface, a first and second face, opposing faces or surfaces, opposing sides that are offset from each other, etc.). In some embodiments, the manual discharge member 510 has a generally square or rectangular shape with rounded corners. In some embodiments, the manual discharge member 510 has a circular or elliptical shape. The manual discharge member 510 includes a protrusion 514 (e.g., an extension, geometry that extends outwards, etc.) that extends from the second side 520, according to some embodiments. The protrusion 514 may extend a distance from the second side 520. In some embodiments, the protrusion 514 is positioned centrally on the second side 520. The protrusion 514 may be configured to engage an internal member of the battery 64 in order to initiate discharge of the cells 504. When the manual discharge member 510 is in the first position or orientation, the first side 518 of the manual discharge member 510 faces or contacts the housing 502 and the protrusion 514 faces outwards. When the manual discharge member 510 is in the second position or orientation, the second side 520 of the manual discharge member 510 faces or contacts the housing 502 such that the protrusion 514 extends through the opening 524 and is received within the inner volume 508 of the housing 502. In some embodiments, the manual discharge member 510 may be maintained in the first position over a lifetime of the battery 64, and when complete discharge or manual discharge of remaining energy in the cells 504 is desired (e.g., at an end of the life of the battery 64), the manual discharge member 510 may be flipped over such that the protrusion 514 extends into the inner volume 508 of the housing 502 (e.g., the second position or orientation) in order to initiate the discharge or depletion of the remaining energy in the cells 504 for safe handling of the battery 64.

Referring particularly to FIGS. 10A-10B and 11A-11B, the manual discharge member 510 is shown in the first position (shown in FIG. 10A) and the second position (shown in FIG. 10B), according to some embodiments. When the manual discharge member 510 is in the first position as shown in FIG. 10A, the protrusion 514 extends outwards from the housing 502. As shown in FIGS. 10A and 10B, the battery 64 includes a seal 546 (e.g., an O-ring, a flexible member, a sealing member, etc.) positioned between the manual discharge member 510 and the housing 502. The seal 546 facilitates sealing between the manual discharge member 510 and the housing 502 when the manual discharge member 510 is in the first position or the second position. In some embodiments, the seal 546 is compressed between the manual discharge member 510 and the housing 502 when the manual discharge member 510 is installed and fastened onto housing 502. The seal 546 facilitates maintaining integrity of the inner volume 508 and reduces a likelihood that moisture may leak into the inner volume 508 of the housing 502.

Referring still to FIGS. 10A and 10B, the battery 64 includes a positive internal terminal 542 of the cells 504 (e.g., a cathode) and a negative internal terminal 544 (e.g., a ground). In some embodiments, the battery 64 includes a conductor 528 (e.g., an electrically conductive element) that

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is coupled with (e.g., directly contacts with) the positive internal terminal 542 such that the conductor 528 can define an electrical energy flow path (e.g., a discharge path). The battery 64 also includes a wire 530 (e.g., a cable, a cord, a copper wire, a conductive element, etc.) including a resistor 532. The battery 64 also includes an adjustable conductive assembly including a conductor 536, an insulator 534, and a conductive tip 538 (e.g., a conductive protrusion, a bump, etc.). The wire 530 may electrically couple with the conductor 536. In some embodiments, the wire 530 is sandwiched between the insulator 534 and the conductor 536. The insulator 534 may be provided on a first side of the conductor 536 that faces the opening 524. The conductive tip 538 may be provided on a second side of the conductor 536 proximate or facing the negative internal terminal 544 (e.g., the ground, the anode or negative terminal of the battery cells 504). In some embodiments, the insulator 534 and the conductor 536 are structurally secured relative to the housing 502 of the battery 64 or relative to the cells 504. For example, the insulator 534 and the conductor 536 may be fastened, secured, fixed, or rotatably coupled with the housing 502 at a first end 552 (e.g., a fixed end). In some embodiments, the insulator 534, the conductor 536, and the conductive tip 538 have the form of a cantilever beam including a fixed and a free end.

Referring to FIGS. 10A-10B, the manual discharge member 510 may be transitionable between the first position or orientation, shown in FIG. 10A, and the second position or orientation, shown in FIG. 10B. When the manual discharge member 510 is transitioned into the second position or orientation, shown in FIG. 10B, the protrusion 514 of the manual discharge member 510 is configured to contact, abut, or directly engage the insulator 534 and thereby drive deflection or movement of the conductor 536 such that the conductive tip 538 contacts, abuts, directly engages, etc., the negative internal terminal 544. When the conductive tip 538 contacts the negative internal terminal 544, electrical current flows between the positive internal terminal 542 and the negative internal terminal 544, through the conductor 528, the resistor 532, the wire 530, the conductor 536, and the conductive tip 538. In this way, the cells 504 may discharge any remaining electrical energy (e.g., residual energy) until the cells 504 are completely depleted of energy and the battery 64 can be properly disposed of. In some embodiments, the conductive tip 538 may be bias into or out of engagement with the negative internal terminal 544 by transitioning the manual discharge member 510 between the first position or orientation (shown in FIG. 10A) and the second position or orientation (shown in FIG. 10B). In this way, the manual discharge member 510 may be transitionable between the first position or orientation and the second position or orientation in order to define an electrical flow path for discharge of the cells 504 (e.g., to permanently and completely discharge the cells 504). In some embodiments, the energy is dissipated at least partially as heat generated by transferring the remaining electrical energy through the resistor 532.

Referring to FIGS. 11A and 11B, the battery 64 may also include a spacer 550 (e.g., an insulator, a rubber member, a ring, a foam member, etc.). The spacer 550 may have the form of a cylindrical or square member including a central opening within which the conductive tip 538 is received. In some embodiments, the spacer 550 is positioned on top of the negative internal terminal 544. The spacer 550 may reduce a likelihood that the conductive tip 538 may accidentally engage or contact the negative internal terminal 544 (e.g., accidental shorting or discharge). In some embodi-

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ments, the spacer **550** is configured to compress and allow the conductive tip to engage the negative internal terminal **544** when the manual discharge member **510** is transitioned from the first orientation to the second orientation as shown in FIGS. **12A-12B**.

Referring to FIG. **12**, a flow diagram of a process **600** for manually discharging or completely depleting a battery of a lift device or other electrical machine includes steps **602-608**, according to some embodiments. In some embodiments, the process **600** is a method for decommissioning a battery at an end of its useful life. In some embodiments, steps **602-608** can be performed to use the battery for one or more functions (e.g., to power an electrical load such as lighting devices, electric motors, electric linear actuators, etc.) during a lifetime of the battery, and to permanently discharge (e.g., completely discharge or deplete cells of the battery) the battery at an end of a life of the battery. Advantageously, the process **600** facilitates improved handling and disposal of batteries that have reached an end of their useful life.

The process **600** includes providing a lift device including a removable battery, the removable battery including a plate configured to be installed over an opening in a first position (e.g., an open state) in which a discharge electrical path is limited and a second position in which a discharge electrical path is defined through a resistor (step **602**), according to some embodiments. In some embodiments, the lift device is the lift device **10** and the removable battery is one of the batteries **64** as described in greater detail above with reference to FIGS. **1-11B**. In some embodiments, the battery is a quick connect or disconnect battery. In other embodiments, the battery is fastened or secured on the lift device and is intended only to be removed during servicing operations of the lift device. The plate may be the manual discharge member **510** which is coupleable (e.g., fastenable) on the housing **502** of the removable battery in either the first position or orientation or the second position or orientation as described in greater detail above with reference to FIGS. **10A-10B** and **11A-11B**.

The process **600** includes performing an operation for the lift device using energy stored in the removable battery while the plate is in the first position (step **604**), according to some embodiments. In some embodiments, step **604** is performed repeatedly over a lifetime of the removable battery. Step **604** may include any charging or discharging operations of the removable battery. For example, the battery may discharge electrical energy to one or more electrical loads of the lift device **10** to perform driving operations, lifting operations, lighting operations, leveling operations, etc. The battery may be charged at appropriate intervals. Over a lifetime of the removable battery, one or more of the cells that store, receive, and discharge energy may degrade, thereby reducing an effectiveness of the removable battery (e.g., degradation due to usage of the cells). In some embodiments, step **604** is performed by the electric motor **52**, the actuators **34**, or any other electrical components of the lift device **10** using energy provided by the removable battery.

The process **600** includes, at an end of a life of the removable battery, removing the removable battery from the lift device (step **606**) and removing the plate from the removable battery, and re-installing the plate over the opening of the battery in the second position such that the discharge electrical path is defined through the resistor such that the cells of the removable battery completely discharge any remaining energy (step **608**), according to some embodiments. In some embodiments, steps **606** and **608** are per-

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formed by a technician. When the technician removes the plate and re-installs the plate in the second position, a positive terminal of the cells of the removable battery may be grounded through a resistor such that any remaining energy in the cells (e.g., residual energy) is completely depleted and at least partially converted into heat by the resistor. The discharge electrical path can be defined using any configuration of components as described in greater detail above with reference to FIGS. **10A-10B** and **11A-11B**.

It should be understood that while the process **600** as described herein is described as being useful at an end of a life of a battery, the process **600** can also be performed for any battery that is to be transported (e.g., damaged batteries, defective batteries, end-of-life batteries, or any other battery that should be fully discharged). Advantageously, the techniques described herein with reference to FIGS. **5-12** facilitate complete discharge of batteries of battery packs that may have a high voltage (e.g., 300 volts or greater across the terminals). The opening **524** in the housing **502** of the battery **64** may also provide access to measure cell stack voltage with a volt meter in order to confirm that the cells **504** of the battery **64** have been fully discharged or depleted.

Referring to FIG. **13**, the conductor **536** or the conductive tip **538** may otherwise be driven into engagement with the negative internal terminal **544** such as by a screw **560** (e.g., a grub screw) that is received through an opening **562** in the housing **502**. In some embodiments, the screw **560** is sub-flush with an exterior surface of the housing **502**. The screw **560** may be threadingly coupled with an inner surface of the housing **502** such that the screw **560** can be driven (e.g., by a technician) to engage the insulator **534** and translate, bend, rotate, or deform the conductor **536** into engagement with the negative internal terminal **544** (e.g., at the conductive tip **538**). In some embodiments, the conductor **536** is otherwise driven to contact or electrically couple with the negative internal terminal **544** in order to discharge or deplete the cells **504**. Advantageously, the systems and methods described herein with reference to FIGS. **5-14** facilitate an internal mechanism for discharging the cells **504** of the battery **64**.

Configuration of the Exemplary Embodiments

As utilized herein with respect to numerical ranges, the terms “approximately,” “about,” “substantially,” and similar terms generally mean $\pm 10\%$ of the disclosed values. When the terms “approximately,” “about,” “substantially,” and similar terms are applied to a structural feature (e.g., to describe its shape, size, orientation, direction, etc.), these terms are meant to cover minor variations in structure that may result from, for example, the manufacturing or assembly process and are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent

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or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having

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machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure.

It is important to note that the construction and arrangement of the lift device 10 and battery 64 as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. It should be appreciated that elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

What is claimed is:

1. A battery pack for a lift device, the battery pack comprising:
 - a housing comprising an opening;
 - a plurality of battery cells positioned within the housing;
 - a resistor electrically coupled with a positive terminal of the plurality of battery cells and positioned within the housing;
 - a conductive element positioned within the housing and configured to transition between an open state in which a discharge path is not defined between the positive terminal of the plurality of battery cells and a negative terminal or a ground, and a closed state in which the discharge path is defined between the positive terminal of the plurality of battery cells to the negative terminal or the ground through the resistor; and
 - a member disposed at the opening and accessible from an exterior of the housing, the member manually transitionable by a technician between a first state in which the conductive element is in the open state, and a second state in which the conductive element is driven into the closed state such that the plurality of battery cells discharge remaining electrical energy via the discharge path;
- wherein the member comprises:
 - a plate having a protrusion in a center, wherein in the first state the member is fastened over the opening such that the protrusion is external to the housing, and wherein in the second state the member is fastened over the opening such that the protrusion

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extends into the housing through the opening and drives the conductive element into the closed state; or

a screw configured to be received within the opening, the screw configured to be accessed from the exterior of the housing by the technician such that the screw is driven to rotate to translate into the second state to bias the conductive element into the closed state.

2. The battery pack of claim 1, wherein the conductive element comprises a cantilever beam including a fixed end and a free end, the conductive element configured to be driven by the member to bend such that a protrusion of the free end engages the negative terminal or the ground in the closed state.

3. The battery pack of claim 1, wherein the battery pack comprises an insulator disposed on a side of the conductive element opposite a side from which a protrusion extends, the member configured to engage the insulator to transition the conductive element from the open state to the closed state.

4. The battery pack of claim 1, further comprising a spacer disposed on a tip of the conductive element, the spacer configured to align the tip of the conductive element and deform as the conductive element is driven by the member to the closed state.

5. The battery pack of claim 1, wherein the conductive element comprises a cantilever beam including a fixed end coupled with the housing and a free end configured to be driven to engage the negative terminal or the ground in the closed state.

6. A lift device, comprising:

a lift assembly configured to raise or lower;

a battery pack configured to provide electrical energy to the lift assembly, the battery pack comprising:

a housing comprising an opening;

a plurality of battery cells positioned within the housing;

a resistor electrically coupled with a positive terminal of the plurality of battery cells and positioned within the housing;

a conductive element positioned within the housing and configured to transition between an open state in which a discharge path is not defined between the positive terminal of the plurality of battery cells and a negative terminal or a ground, and a closed state in which the discharge path is defined between the positive terminal of the plurality of battery cells to the negative terminal or the ground through the resistor, wherein the conductive element comprises a cantilever beam including a fixed end coupled with the housing and a free end configured to be driven to engage the negative terminal or the ground in the closed state; and

a member disposed at the opening and accessible from an exterior of the housing, the member manually transitionable by a technician between a first state in which the conductive element is in the open state, and a second state in which the conductive element is driven into the closed state such that the plurality of battery cells discharge remaining electrical energy via the discharge path.

7. The lift device of claim 6, wherein the member comprises a plate having a protrusion in a center, wherein in the first state the member is fastened over the opening such that

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the protrusion is external to the housing, and wherein in the second state the member is fastened over the opening such that the protrusion extends into the housing through the opening and drives the conductive element into the closed state.

8. The lift device of claim 6, wherein the member comprises a screw configured to be received within the opening, the screw configured to be accessed from the exterior of the housing by the technician such that the screw is driven to rotate to translate into the second state to bias the conductive element into the closed state.

9. The lift device of claim 6, wherein the conductive element comprises a cantilever beam including a fixed end and a free end, the conductive element configured to be driven by the member to bend such that a protrusion of the free end engages the negative terminal or the ground in the closed state.

10. The lift device of claim 6, wherein the battery pack comprises an insulator disposed on a side of the conductive element opposite a side from which a protrusion extends, the member configured to engage the insulator to transition the conductive element from the open state to the closed state.

11. The lift device of claim 6, further comprising a spacer disposed on a tip of the conductive element, the spacer configured to align the tip of the conductive element and deform as the conductive element is driven by the member to the closed state.

12. The lift device of claim 6, wherein the lift device is a fully electric boom.

13. A battery pack for a lift device, the battery pack comprising:

a housing comprising an opening;

a plurality of battery cells positioned within the housing;

a resistor electrically coupled with a positive terminal of the plurality of battery cells and positioned within the housing;

a conductive element positioned within the housing and configured to transition between an open state in which a discharge path is not defined between the positive terminal of the plurality of battery cells and a negative terminal or a ground, and a closed state in which the discharge path is defined between the positive terminal of the plurality of battery cells to the negative terminal or the ground through the resistor; and

a member disposed at the opening and accessible from an exterior of the housing, the member manually transitionable by a technician between a first state in which the conductive element is in the open state, and a second state in which the conductive element is driven into the closed state such that the plurality of battery cells discharge remaining electrical energy via the discharge path;

wherein the battery pack further comprises at least one of:

(i) an insulator disposed on a side of the conductive element opposite a side from which a protrusion extends, the member configured to engage the insulator to transition the conductive element from the open state to the closed state; or

(ii) a spacer disposed on a tip of the conductive element, the spacer configured to align the tip of the conductive element and deform as the conductive element is driven by the member to the closed state.

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